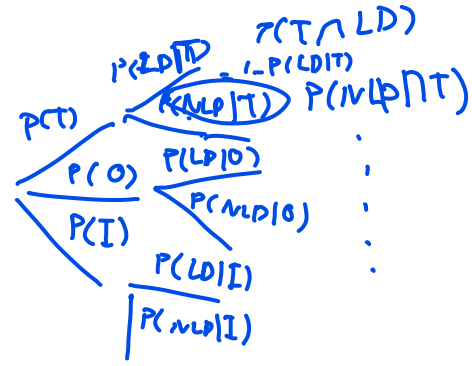
	LD	ND
T	$P(T \cap LD)$	
O	$P(O \cap LD)$	
I	$P(I \cap LD)$	
	$P(LD)$	$P(ND)$

$P(T \cap ND) = (1 - P(LD|T)) \cdot P(T)$



Worksheet 5

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2022-10-10

$P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

$P(A \cap B) = P(A|B)\,P(B)$

1. 60% of the student body at UTC is from the state of Tennessee (T), 30% are from other states (O), and the remainder constitutes international students (I). Twenty percent of students from Tennessee lives in the dormitories, whereas, 50% of students from other states live in the dormitories. Finally, 80% of the international students live in the dormitories. $P(LD|O)$

   (a) What percentage of UTC students lives in the dormitories? $= T\ I$

   (b) Given that a student lives in the dormitory, what is the probability that they are an international student?

   (c) Given that a student does not live in the dormitory, what is the probability that they are an international student?

$P(T) = 0.6$, $P(O) = 0.3$, $P(I) = 0.1 = 1 - 0.6 - 0.3$

$P(LD|T) = 0.2$ $\quad P(LD|O) = 0.5$, $\quad P(LD|I) = 0.8$

(a) $P(LD) = P(LD \cap T) + P(LD \cap O) + P(LD \cap I)$

$= P(LD|T) \cdot P(T) + P(LD|O)\,P(O) + P(LD|I)\,P(I)$

$= 0.2 \times 0.6 + 0.5 \times 0.3 + 0.8 \times 0.1$

$= 0.12 + 0.15 + 0.08 = 0.35$

b) $P(I \mid LD) = \dfrac{P(I \cap LD)}{P(LD)} = \dfrac{P(LD|I) \cdot P(I)}{P(LD)}$ Bayes Rule

$P(LD^c|I) = \dfrac{P(LD^c \cap I)}{P(I)}$

$P(LD^c|I) \cdot P(I) = P(LD^c \cap I)$

$= \dfrac{0.8 \times 0.1}{0.35} = \dfrac{0.08}{0.35} = 0.229$

c) $P(I \mid (LD)^c) = \dfrac{P(I \cap (LD)^c)}{P(LD^c)} = \dfrac{P((LD)^c|I)\,P(I)}{1 - P(LD)}$ Not living in Dorm

$P(LD^c) = 1 - P(LD)$

$= \dfrac{[1 - P(LD|I)]\,P(I)}{1 - P(LD)}$

$$= \frac{[1 - \overset{0.2}{0.8}]\ 0.1}{1 - 0.35} = \frac{0.02}{0.65} = 0.03$$

2. A survey asked 100 residents in a town whether they are smokers. Given the following information on the residents' response:

Wam

	Daily Workout	No daily workout	Total
Non-smoker	40	30	70
Smoker	20	10	30
Total	60	40	100

(a) Find the joint probability table.

(b) What is the probability that a randomly selected resident worksout daily? $P(\text{Workout} = \text{Daily}) = P(WD \cap NS) + P(WD \cap S) = 0.6$

$0.4 = \alpha$ (c) What is the probability that a randomly selected resident doesn't workout daily?

$= P(NW \cap S) + P(NW \cap NS)$

(d) What is the probability that a randomly selected resident worksout daily and is a smoker?

(e) What is the probability that a randomly selected resident worksout daily or is a smoker?

(f) A randomly selected resident doesn't workout daily. What is the probability that the resident is a smoker?

(a) dividing table by 100

	Workout	N Workout	
Non-S	$\frac{40}{100} = 0.4$	$\frac{30}{100} = .3$	$\frac{70}{100}$ 0.7
S	$\frac{20}{100} = 0.2$	$\frac{10}{100} = 0.1$	$\frac{30}{100} = 0.3$
	$\frac{60}{100} = 0.6$	$\frac{40}{100} = 0.4$	$\frac{100}{100}$ 1

(e) $P(WD \cup \text{Smoker}) = P(WD) + P(\text{Smoker}) - P(WD \cap S)$

$= 0.6 + 0.3 - 0.2$

$= 0.9 - 0.2 = 0.7$

(f) $P(S \mid NWD) = \frac{P(S \cap NWD)}{P(NWD)} = \frac{0.1}{0.4}$

$= \frac{1}{4} = 0.25$

didn't do

3. The number of pizzas delivered to university students each month is a random variable with the following probability distribution.

X	0	1	2	3 or more
P(x)	0.1	0.3	0.4	0.2

(a) Find $P(X \leq 2)$.

(b) $P(1 \leq X \leq 2)$.

(c) Determine the mean and variance of X.

4. The number of persons living per household in a city was collected and were summarized. The frequency table is given below:

Number of persons	1	2	3	4	5	6	7
Number of households (millions)	3.1	4.5	3.8	2.5	2.4	2.5	1.2

Define X as the number of people per household.

(a) Write down the probability distribution of X.

(b) Find $P[X \geq 4]$

(c) Find $\mathrm{P}[2 \leq \mathrm{X} < 4]$

(d) Find the mean, variance and standard deviation for number of people per household.